

OSBAMS Rev.2 — bench verification V1–V10 (one page, fill by hand)

NO BATTERY is connected for any item. Power OFF unless a row says otherwise (V1 needs U1 powered, V6 coil current needs the 12 V supply). Instruments: EDU34450A (DMM), EDUX1052G (scope). Rows ordered by priority (P1 first); IDs match the audit. Unknown = write UNKNOWN, never guess.

Date _____ Operator _____ 2nd person _____ EDU34450A asset ID _____ cal status _____ Repo commit _____

#	Pri	Item	What to check	Record here	Unlocks
V7	P1	Diode orientation D1/D2/D3	Photo + cathode-band side of each installed diode relative to the net it should hold (D3 band toward the coil/relay side; D2 band to +12 V; D1 TVS band to +12 V). Power OFF; EDU34450A diode test as a cross-check.	D1 band at: _____ D2 band at: _____ D3 band at: _____ D3 Vf fwd ____ V	G-01 BLOCKER: confirms/refutes the symbol-vs-footprint polarity defect
V6	P1	Relay K1, ARM, E-stop block	Relay: FULL marking + date code, coil resistance (expect ~90 ohm), coil current at 12 V. ARM: marking, where it sits in the coil path (before/after E-stop, or unconnected). E-stop block: NC contact count fitted, NO count.	relay P/N _____ coil ____ ohm ____ mA ARM in path: _____ E-stop NC x ____ NO x ____	G-02 relay keep/replace; G-05 PA0 option A (2nd NC) vs B (opto); G-14 ARM
V1	P1	TC74: four discriminators (NO redesign until all four filled)	(1) Package marking A-code/V-code. (2) VDD at pin 5 and idle SDA/SCL levels, U1 powered. (3) I2C1_Scan + raw TC74_Init status (NACK/TIMEOUT/BUS_ERROR/NOT_READY). (4) Scope SCL during a TC74 transaction: tLOW vs 4.7 us, tHIGH vs 4.0 us.	(1) _____ (2) VDD ____ V SDA ____ SCL ____ (3) scan 0x _____ status _____ (4) tLOW ____ us tHIGH ____ us	G-04: root cause before any TC74 interface change
V9	P1	Nucleo <-> J5 wiring, power, host path	Which Nucleo pins go to J5 (PC8, PC9, PA0, PA1?). How the Nucleo is powered (USB / 5V / VIN / E5V). Host link: ST-LINK USB VCP or separate UART (PA2/PA3 solder bridges).	J5 pins: _____ Nucleo power: _____ host path: USB / UART	G-05 / G-18 / G-21: Nucleo vs PCBA scope; Pi link
V2	P1	VO610A feedback stage (K1 -> PC9)	Photograph; record LED-side resistor values, output pull-up (value, to what rail), what the two 1N4148 do (antiparallel? series?), pack-side connection point (relay load side?).	Rled ____ ohm pull-up ____ to ____ 1N4148 role: _____ taps relay: _____	G-05: K1 feedback schematic for PC9
V4	P1	Installed ADC divider (-> PA1)	Resistor values and tolerances; where the tap goes (to PA1 directly? cap?); where the ground return connects; pack-side connection point. Power OFF.	Rtop ____ tol ____ Rbot ____ tol ____ tap -> _____ GND return -> _____	G-05: protected divider; 16:1 firmware scale
V3	P2	INA228 module	Pin order vs J4 (3V3/GND/SCL/SDA), ALERT/A0/A1 straps, onboard shunt present/removed, how IN+ / IN- / VBUS are wired to the RSA-20-50.	pin order: _____ A0/A1: ____ / ____ onboard shunt: Y/N IN+/IN-/VBUS: _____	G-12: Kelvin interface; K3 module kept
V10	P2	Ground-bond point	Single point where pack-negative meets logic GND? Any second path (USB shield, scope ground, Pi)? Continuity pack(-) to GND with power OFF.	bond at: _____ 2nd path? Y/N R pack(-)-GND: _____	G-13: ground bond
V8	P2	R1/R2/C2/C3 + connectors	Installed R1, R2, C2, C3 values and package; terminal-block and header part numbers (J1, J2, J3, J4, J5, J6) from markings/purchase records.	R1 ____ R2 ____ C2 ____ C3 ____ terminals: _____ headers: _____	G-07 / G-08: exact MPNs
V5	P3	Fuse, holder, pack-bus TVS	Holder marking and its datasheet DC V / A rating; installed fuse marking (expect 15 A, DC rating); any TVS across the pack bus.	holder _____ VDC ____ A fuse _____ pack TVS? Y/N _____	G-15 / G-16

Done when every row is filled (or UNKNOWN). Then: update **REV1_HARDWARE_TRUTH_INVENTORY.md** → set **rev1_physical_observations_recorded** in RELEASE_GATES.json only with these sheets filed → revise the existing KiCad board. P1 = blocks schematic decisions (BLOCKER-linked); P2 = HIGH, needed before release; P3 = MEDIUM.

Signature _____ Sheet stored at _____